

CSE-251

ELECTRONIC CIRCUITS

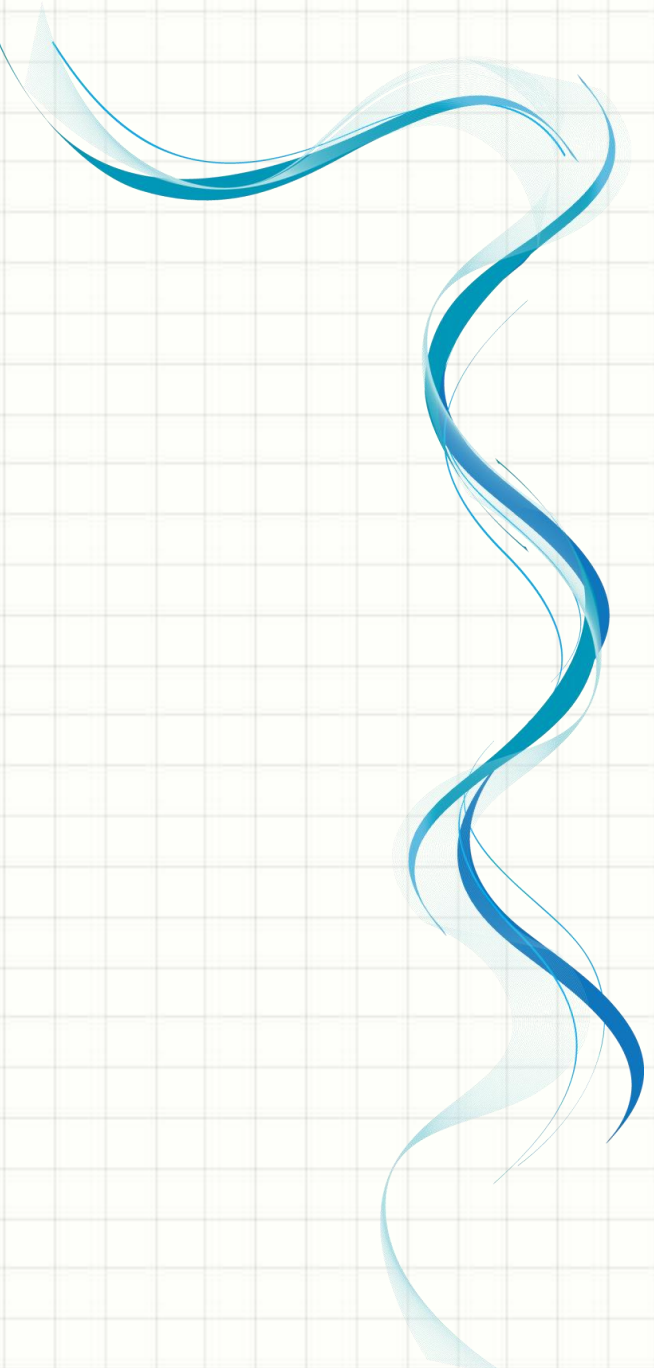
LECTURE-03

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Today's Overview:

- ☐ *Rutherford's Model*
- ☐ *Bohr's Model*
- ☐ *Energy Levels and Energy Bands*
- ☐ *Different Types of Materials*



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Rutherford's Model

Rutherford's atomic model:

Ernest Rutherford, a British scientist conducted alpha particle scattering experiment in **1909** and based on the observations of this experiment, he explained the atomic structure of elements and proposed Rutherford's Atomic Model in **1911**.

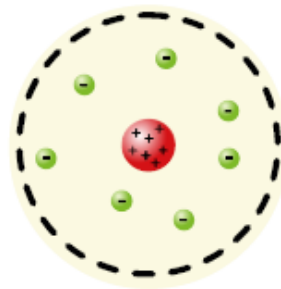


"Of all created comforts, God is the lender; you are the borrower, not the owner."

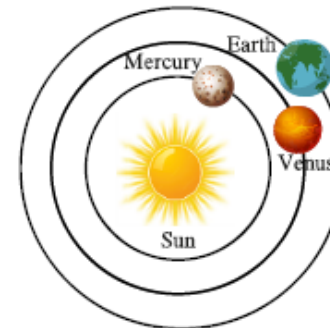
Ernest Rutherford

Rutherford's postulates:

- The positive charge and most of the mass of an atom is concentrated in an extremely small volume called **nucleus**.
- The negatively charged electrons surround the nucleus of an atom. The electrons surrounding the nucleus revolve around it with very high speed in circular paths. He called **orbits**.
- Electrons being negatively charged and nucleus being a densely concentrated mass of positively charged particles are held together by a strong **electrostatic force of attraction**.



Rutherford's Atomic Model



Solar Model

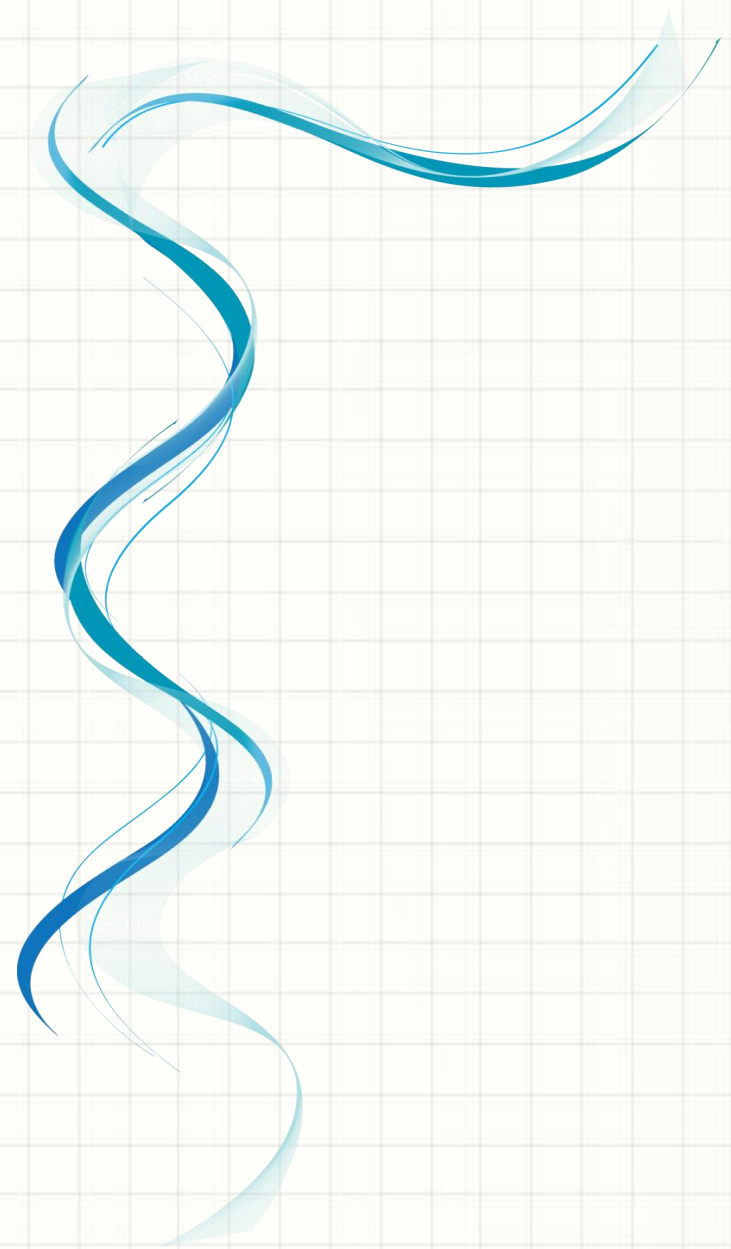
Limitations:

- Rutherford used the solar model as the basis for his atomic model. The planets in the solar system revolve around the sun due to the force of gravity. However, in an atom, the electron is negatively charged and the nucleus is positively charged. There is also an electric force between them. The behavior of electric force and gravitational force is not the same. Rutherford could not explain how the electron is revolving around.
- According to Rutherford's atomic model, the electrons in an atom revolve around the positively charged nucleus. However, there is no statement in Rutherford's atomic model about how they revolve in which orbit. In other words, Rutherford's atomic model does not provide any idea about the shape of the orbits. Rutherford did not give any clear idea about whether one or more electrons will be in the same orbit or in different orbits in his atomic model.

Limitations:

- According to Maxwell's electromagnetic theory, a charged particle that is accelerated radiates energy. Since the electron has centripetal acceleration during rotation, it is natural for the electron to radiate energy continuously. As a result, the radius of the electron's orbit will gradually decrease as the energy decreases. The orbit of the revolving electron will become spiral, and at a certain time the electron will fall into the nucleus. Therefore, there can be no permanent existence of an atom of this type. Hence, the stability of the atom is questioned according to Rutherford's atomic model.

Bohr's Model



Bohr's atomic model:

The Bohr model of the atom was proposed by Niels Bohr in 1915. It came into existence with the modification of Rutherford's model of an atom. Niels Bohr was awarded the Nobel Prize in physics in 1922 for his work investigating the structure of an atom.



"An expert is a person who has made all the mistakes that can be made in a very narrow field."

- Niels Bohr

Bohr's postulates:

- In an atom, negatively charged electrons revolve around the positively charged nucleus in a definite circular path called **Orbits** or **Shells**. Each orbit or shell has a fixed energy and a fixed angular momentum. Electronic distribution of various orbits or energy levels can be calculated by the formula $2n^2$. Here, 'n' denotes the **number of orbits**.

For example: The number of electrons in K shell (1st orbit) can be calculated by $2n^2 = 2 \times 1^2 = 2$. Thus, maximum number of electrons in 1st orbit = 2.

Similarly, The number of electrons in L shell (2nd orbit) = $2 \times 2^2 = 8$. Thus, maximum number of electrons in 2nd orbit = 8.

Angular momentum: $mvr = \ell = \frac{nh}{2\pi}$

where-

ℓ = angular momentum

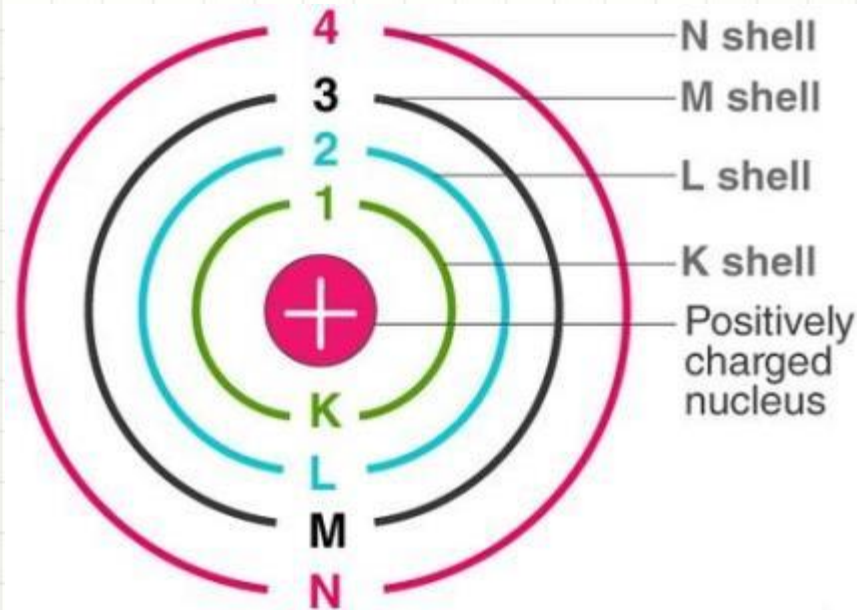
n = principal quantum number

h = Planck's constant

π = pi

Bohr's postulates:

- The energy levels are represented by an integer ($n=1, 2, 3\dots$) known as the **Quantum number**. The range of quantum number starts from nucleus side with $n=1$ having the lowest energy level. The orbits $n=1, 2, 3, 4\dots$ are assigned as K, L, M, N.... shells and when an electron attains the lowest energy level, it is said to be in the **Ground state**.

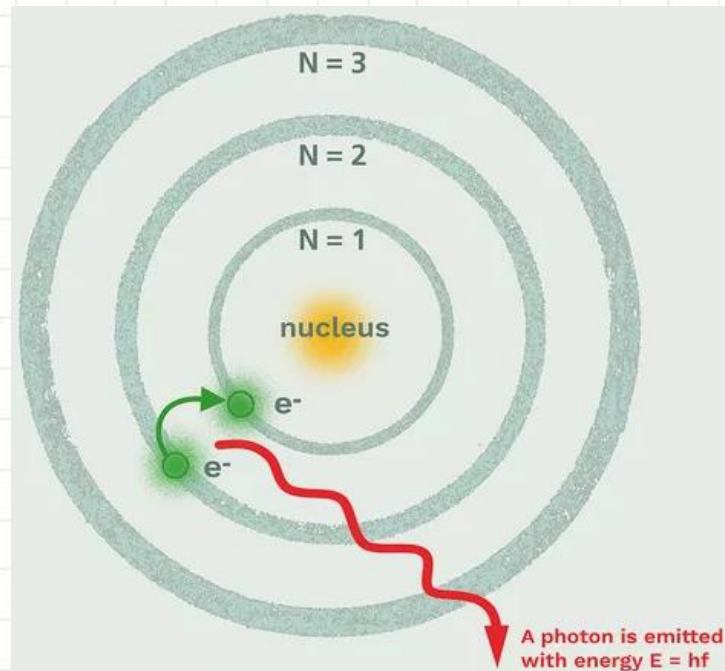


Bohr's Model of an Atom

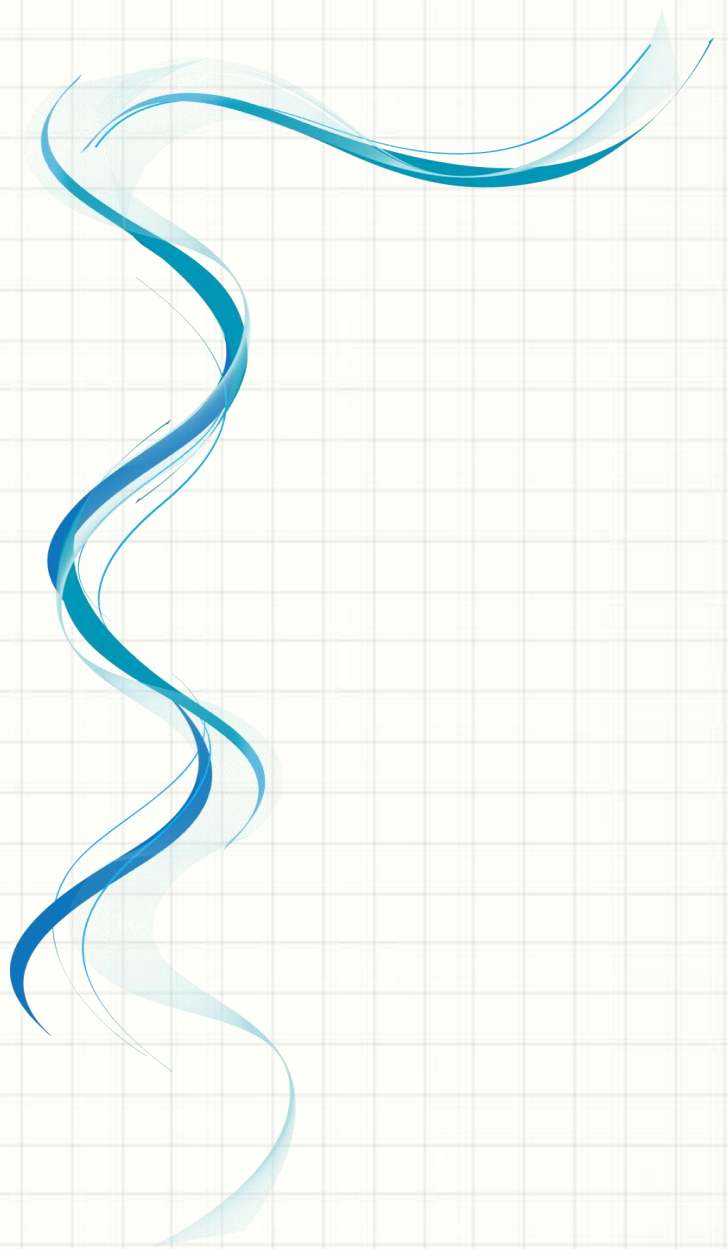
Bohr's postulates:

- The electrons in an atom move from a lower energy level to a higher energy level by gaining the required energy and an electron moves from a higher energy level to lower energy level by losing energy as photons.

$$\Delta E = E_2 - E_1 = h\nu, \text{ where } h \text{ is Planck's constant.}$$

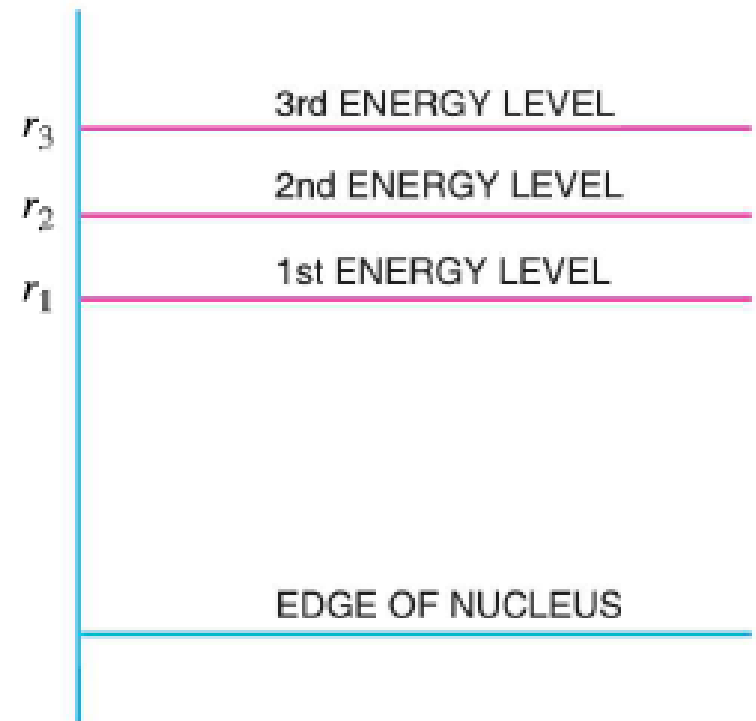
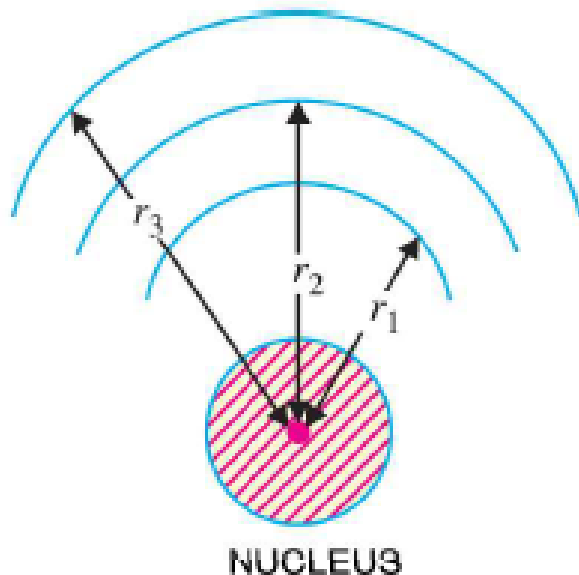


Energy levels and Energy Bands



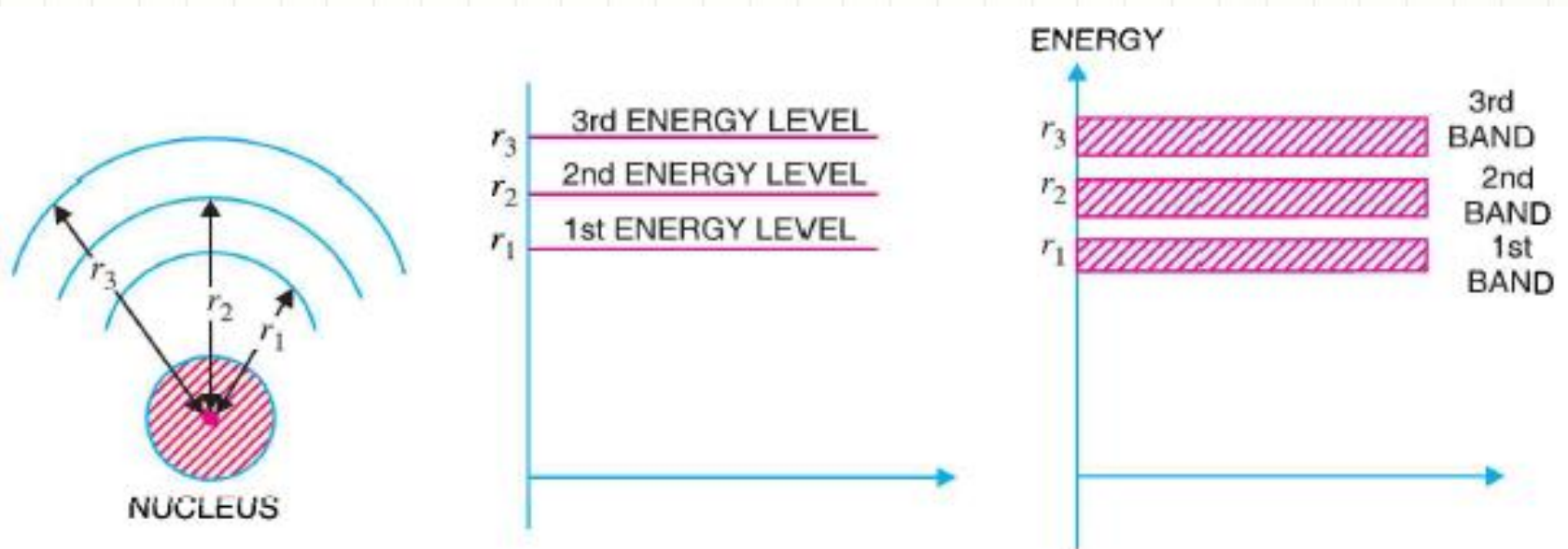
Energy Levels:

The electrons moving in a particular orbit possess the energy of that orbit. The larger the orbit, the greater is its energy.



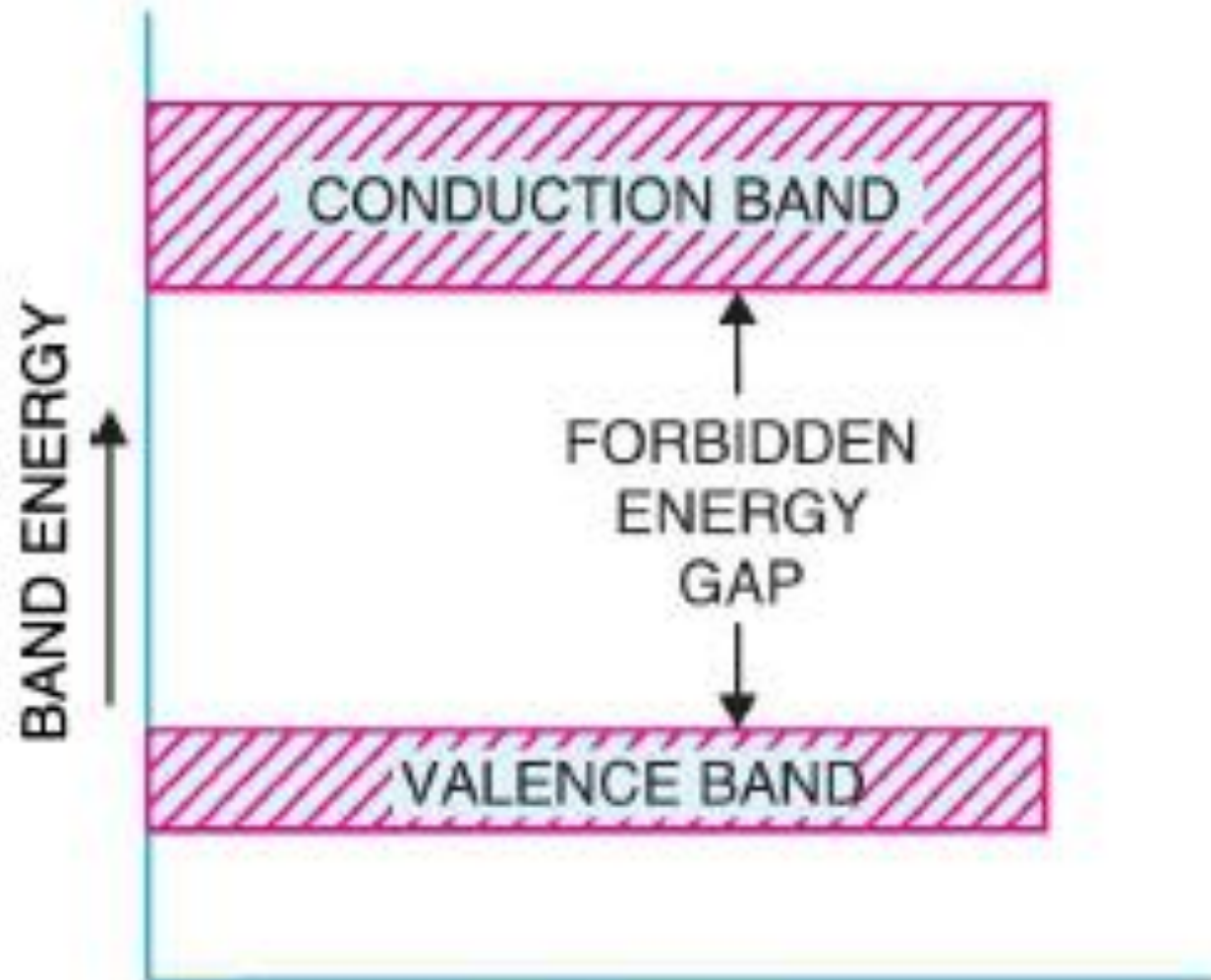
Energy Bands:

The range of energies possessed by an electron in a solid is known as energy bands.



In case of a single isolated atom, the electrons in any orbit possess definite energy. However, an atom in a solid is greatly influenced by the closely-packed neighbouring atoms. The result is that the electron in any orbit of such an atom can have a range of energies rather than a single energy. This is known as *energy band*.

Energy Band Diagram



Energy Band Diagram

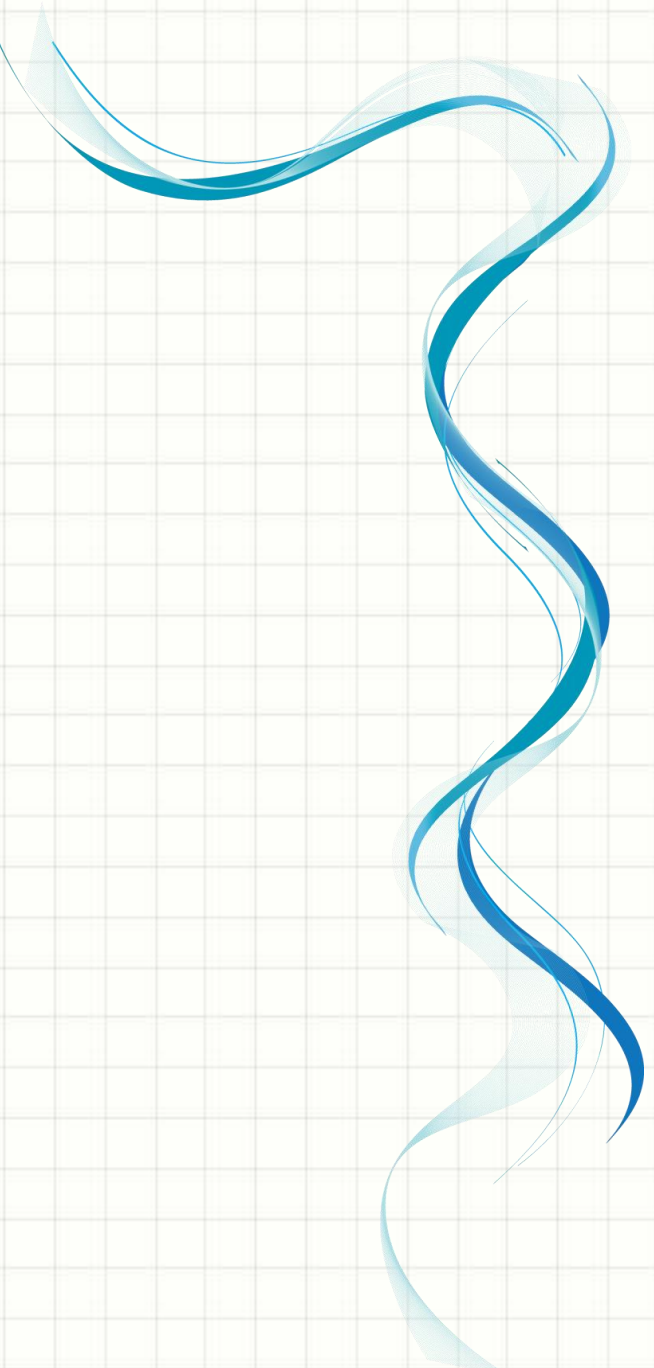
(i) **Valence band.** *The range of energies (i.e. band) possessed by valence electrons is known as **valence band**.*

(ii) **Conduction band.** In certain materials (*e.g.* metals), the valence electrons are loosely attached to the nucleus. Even at ordinary temperature, some of the valence electrons may get detached to become free electrons. In fact, it is these free electrons which are responsible for the conduction of current in a conductor. For this reason, they are called *conduction electrons*.

*The range of energies (i.e. band) possessed by conduction band electrons is known as **conduction band**.*

(iii) **Forbidden energy gap.** *The separation between conduction band and valence band on the energy level diagram is known as **forbidden energy gap**.*

No electron of a solid can stay in a forbidden energy gap as there is no allowed energy state in this region. The width of the forbidden energy gap is a measure of the bondage of valence electrons to the atom. The greater the energy gap, more tightly the valence electrons are bound to the nucleus. In order to push an electron from valence band to the conduction band (*i.e.* to make the valence electron free), external energy equal to the forbidden energy gap must be supplied.

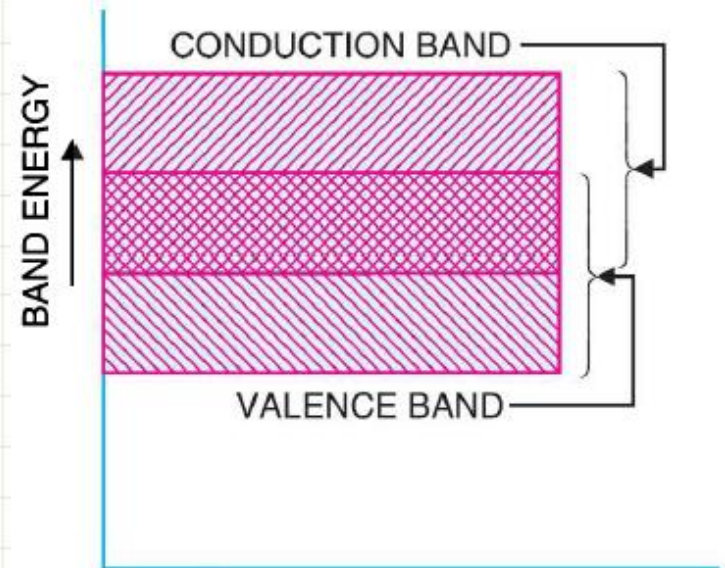
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Different types of Materials

Conductors:

Examples- Copper, Silver, Aluminium, Tungsten etc.

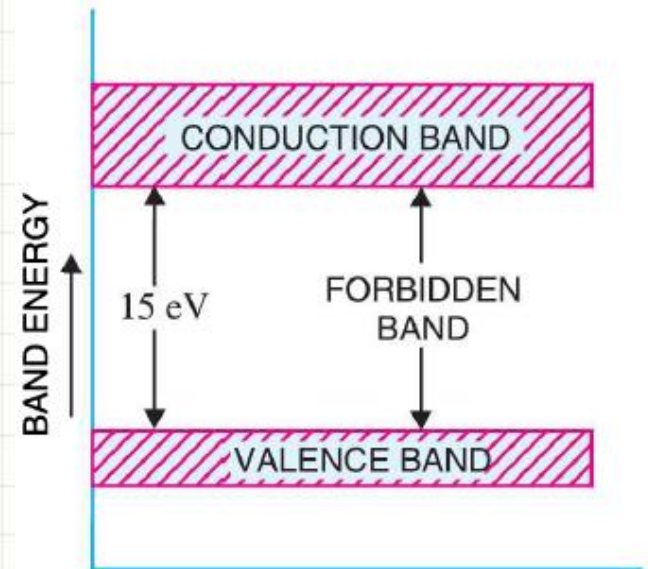
- ❑ *Electrical current easily passes through conductors, as large number of free electrons are available.*
- ❑ *The valance band and conduction band overlap each other.*
- ❑ *The forbidden energy gap = 0 eV.*
- ❑ *Positive temperature coefficient of resistance.*



Insulators:

Examples- Glass, Rubber, Wood etc.

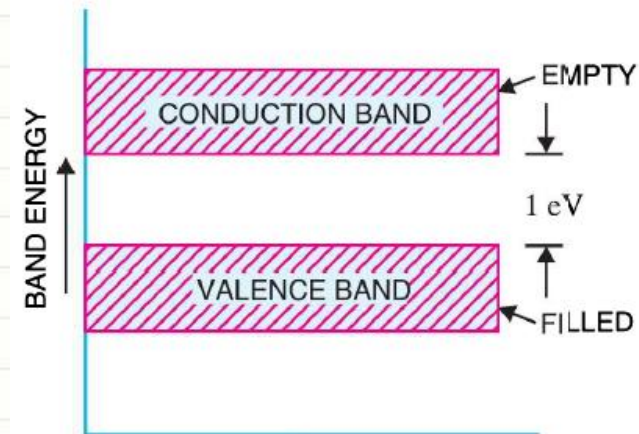
- ❑ *Electrical current does not pass through insulators.*
- ❑ *The valance band is full, whereas the conduction band is completely empty.*
- ❑ *The forbidden energy gap is large enough (≈ 15 eV).*
- ❑ *The electrons of the valance band do not have enough energy to cross and move towards conduction band, therefore the conductivity of the insulator is almost zero.*
- ❑ *Negative temperature coefficient of resistance*



Semiconductors:

Examples- Germanium, Silicon etc.

- ❑ *Small electrical current passes easily through semiconductors.*
- ❑ *The conductivity of semiconductor lies between insulator and conductor.*
- ❑ *The valance electron is completely filled whereas the conduction band is almost empty.*
- ❑ *The forbidden energy gap is very small (≈ 1 eV).*
- ❑ *The semiconductor behaves as an insulator at room temperature or low temperature.*
- ❑ *Negative temperature coefficient of resistance.*





QUESTIONS?

SEE YOU IN THE NEXT CLASS

